Chenxu Liu 刘宸旭

B.Sc. in Physics — Shandong University, Qingdao

202422201069@mail.sdu.edu.cn · physicsliu299792458@outlook.com · github.com/PhyLiusir
Qingdao, China

EDUCATION

Shandong University, Qingdao

2024 – 2028

Institute of Frontier and Interdisciplinary Science · class of 2024, expected 2028

Physics for the foundations. Coursework: theoretical mechanics, electrodynamics, thermodynamics and statistical physics, quantum mechanics, advanced quantum mechanics, quantum field theory, particle physics, general relativity. Time outside it goes to engineering on both sides of large models and agents — read, reproduce, then change it on purpose.

RESEARCH INTERESTS

Large models: deployment and serving economics, post-training (SFT → preference → RL), and new attention architectures — Loop Attention.
Agents: agent construction, tool design, multi-agent collaboration, retrieval-augmented generation.

RESEARCH DIRECTIONS & WORK IN PROGRESS

- **Deployment (LM 01).** Cost structure of one model across serving stacks, batching and precision — cost per token, P99 latency and peak memory read together.
- **Post-training (LM 02).** A full SFT → preference → RL chain reproduced at small scale, then ablated stage by stage; focus on which gains survive on a smaller model.
- **Loop Attention (LM 03).** Reusing one attention block across depth — memory, wall-clock and training stability measured against a stacked baseline. Main line.
- **Agent construction (AG 01).** The smallest useful combination of loop, memory and planning; a taxonomy of tool-less agent failures as the starting object.
- **Tool design (AG 02).** Schemas, error surfaces and return shapes for a batch of real tasks, comparing failure distributions before and after.
- **Multi-agent collaboration (AG 03).** A strong single-agent baseline first; multi-agent setups have to beat it reliably before roles get split.
- **RAG (AG 04).** A citation-required QA set and a scoring script: an answer must land on a specific passage or it counts as wrong.
- **Light-front quantization (PH 01).** Fock-space truncation as the one approximation that has to be justified.
- **Baryon light-cone Fock expansion (PH 02).** Three-quark sector first; which observables survive a cut at fixed N.
- **Proton structure (PH 03).** DIS, SIDIS and DY read together — collinear PDF, transverse momentum, 3D distributions.

TECHNICAL SKILLS

LANGUAGES	Python (primary), C++, CUDA, Bash, Git
MODEL SIDE	PyTorch training and fine-tuning, transformers and custom layers, serving and quantisation, LoRA and preference alignment
AGENT SIDE	Tools and function calling, multi-agent orchestration, vector retrieval and RAG, evaluation sets and scoring scripts
TOOLING	Linux and containers, distributed training basics, reproducible experiments

LANGUAGES

Chinese (native) · English (working proficiency).